

Full name: Ha Hoang Phuc

MSSV: 222683

Class: DH22TIN07

LESSON 1: Library management system

- Identify candidate classes (Books, Readers, Vouchers, Employees, Publishers) +

Book: main entity, managing book information.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class +

Book

- Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), manNXB (FK), soLuongTon (int), trangThai (string: "Yes available"/"On loan").
- Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho ().

+ DocGia

- Attributes: maDocGia (PK, string), hoTen (string), birth date (date), diaChi (string), soDienThoai (string), ngayDangKy (date), gioiTinh (boolean).
- Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().

+ PhieuMuon

- Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), FK, you're (date), you're (date), you're (date), TinhTrang (string: "On loan"/"Paid"/"Overdue").
- Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat().

+ NhanVien

- Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), ngaySinh (date), soDienThoai (string).
- Behavior: dangNhap(), quanLyDocGia(), quanLyMuonTra(), quanLySach ().

+ NhaXuatBan

- Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).
- Behavior: addNXB(), capNhatNXB().

- Relationships

 ☐ Books and Publishers: 1 book – 1
publisher

- ☐☐ Readers and Borrowing Slips: 1 reader – many borrowing slips
- ☐☐ Books and borrowing slips: 1 book – many borrowing slips
- ☐☐ Employees and Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ☐ DangKyTaiKhan (Customer registration).
- ☐ DangNhap (Sign in).
- ☐ SeeSanPham (Search / View product list).
- ☐ ThemVaoGioHang (Add to cart).
- ☐ DatHang (Order).
- ☐ ThanhToan (Online payment / COD).
- ☐ View DonHang (View order history).

Admin / NguoiQuanLy:

- ☐ QuanLySanPham (Add/Edit/Delete products).
- ☐ QuanLyDonHang (View/Confirm/Cancel order).
- ☐ QuanLyKhachHang (Customer information management).
- ☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ☐ ThanhToan includes XacNhanDonHang.
- ☐☐ DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- ☐☐ Customer (Customer): from register, view orders.
- ☐☐ SanPham (Product): product management.
- ☐☐ GioHang (Cart): temporarily store products before ordering.
- ☐☐ Don Hang (Order): from order, payment.
- ☐☐ ChiTietDonHang (OrderDetail): product details in the order.
- ☐☐ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- ☐☐ Admin or Personnel (inherited from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)
 - ☑ KhachHang and DonHang: 1 customer – many orders
 - ☑ DonHang and ChiTietDonHang: 1 order – many order details (Composition)
 - ☑ ChiTietDonHang and SanPham: many single details – 1 product
 - ☑ DonHang and ThanhToan: 1 order – 1 payment
 - ☑ User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and Admins are all the same type of User, sharing the same account, password, and full name).

Lesson 3: Convert class diagrams into data tables

1.MonHoc Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	